

TITLE PAGE

Problem Statement ID – SIH25185

Problem Statement Title - AI-Powered DPR Quality

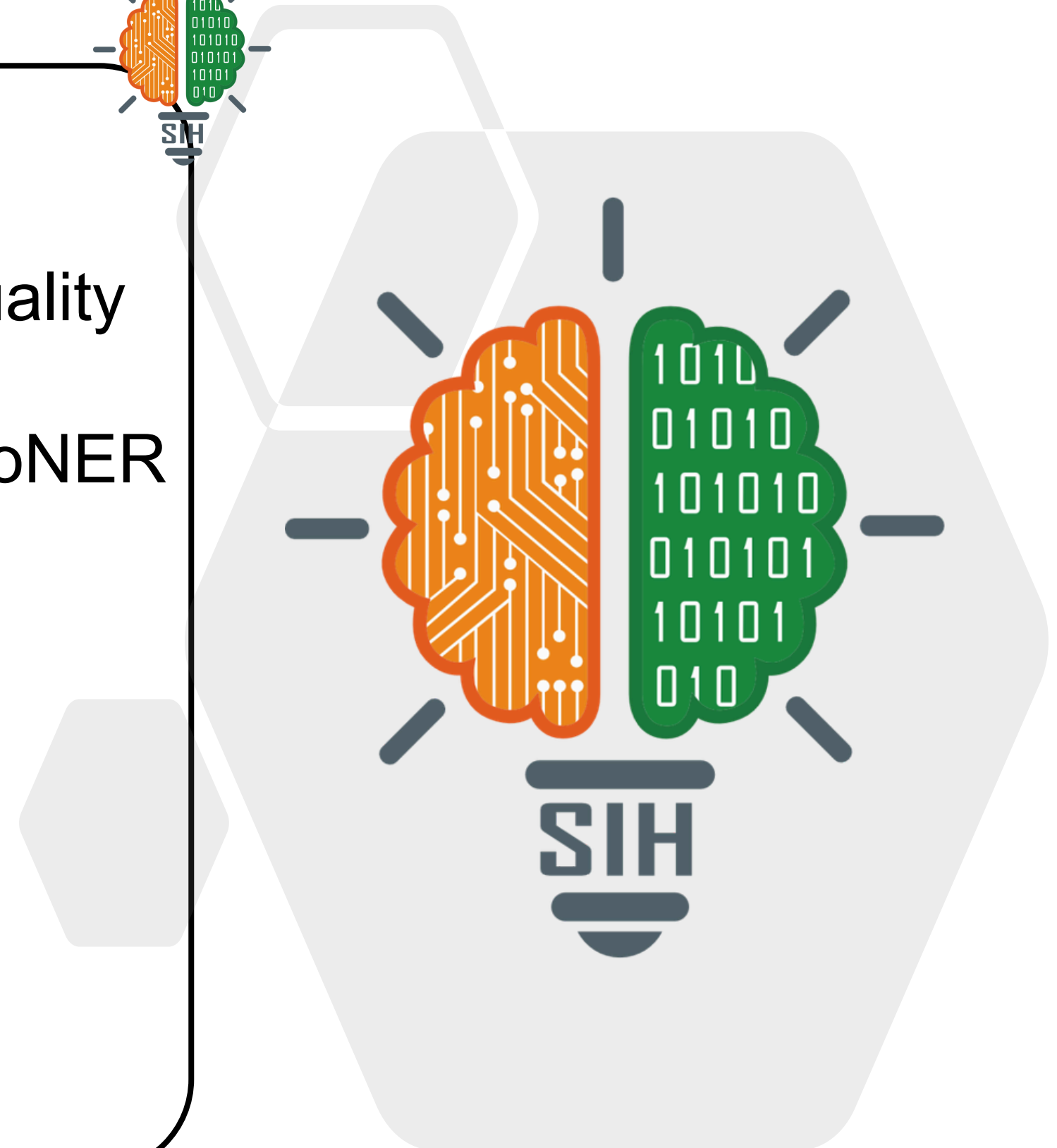
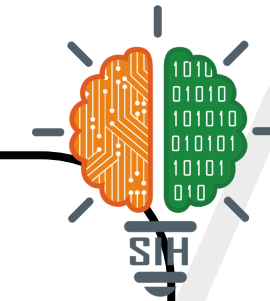
Assessment and Risk Prediction System for MDoNER

Theme - Smart Automation

PS Category - Software

Team ID - 107181

Team Name - __NEXUS__IITB



IDEA TITLE

IDEA/SOLUTION

The aim of this idea is to process large DPRs using a hybrid pipeline that combines **text extraction**, **localized summarization**, and a **domain-specific AI-trained LLM** to generate risk analysis and actionable insights for effective DPR assessment.

IMPLEMENTATION AND FEATURES

1. **End-to-end automated** pipelining: Includes the aforementioned functionalities and displays the analysis via interactive charts and dashboards.
2. **Two-tier LLM framework**: Breaks down large DPRs into processable summaries, processed by a **second-level fine-tuned LLM** for final output.
3. **Well-structured output formatting**: **JSON files** capture project metadata, timelines, **risk likelihood/impact scores**, and a **DPR quality card** for verification.
4. **Interactive Dashboard**: Real-time PDF upload and processing feedback with **progress indicators**, and comprehensive analysis display featuring risk scores, cost breakdowns, and timeline evaluations presented through clear visualizations.
5. **Structured Output Generation**: Automated generation of structured JSON outputs containing project enabling **seamless integration with existing government systems** and facilitating data-driven decision workflows.

PROBLEM RESOLUTION

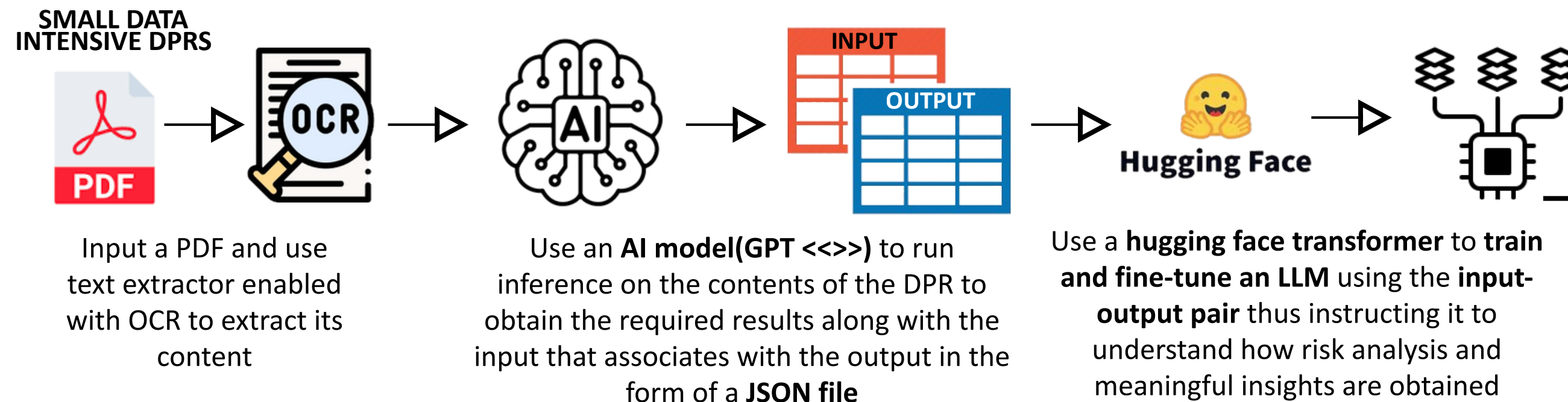
Problem: DPR reviews face **lengthy documents**, **inconsistencies**, **hidden risks**, **weak auditability**, and **sensitive government data**.

Solution: Our system provides **concise summaries**, **automated scoring**, **anomaly validation**, full provenance, and **secure on-premise AI** for efficient, auditable DPR assessment.

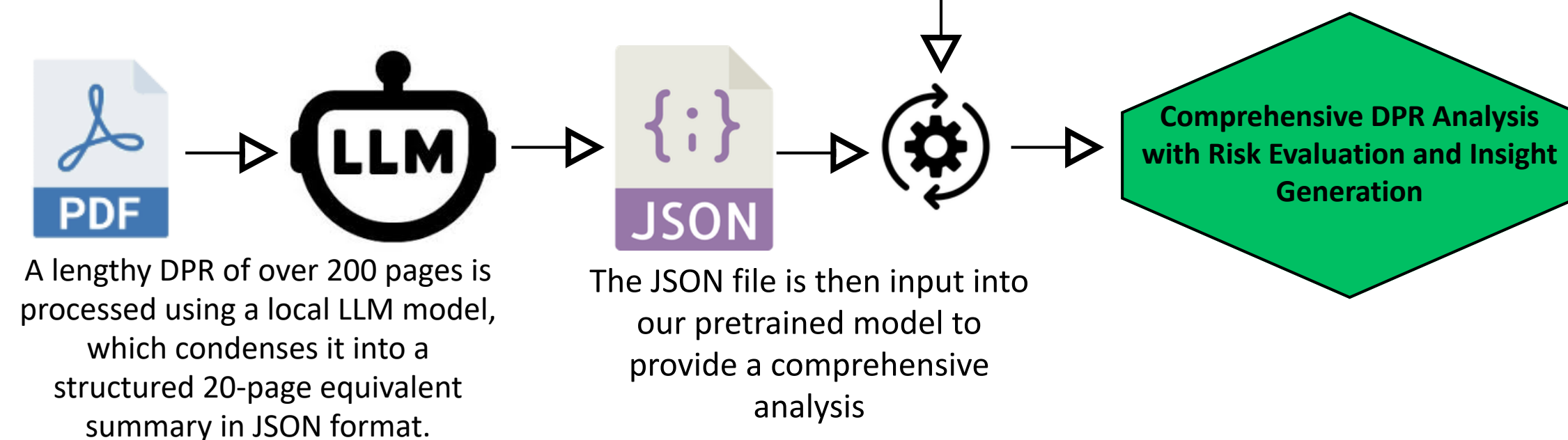
PROPOSITION UNIQUENESS

Our solution uses a **multi-layer LLM pipeline**: one layer condenses lengthy DPRs into **structured summaries**, while a **domain-trained** layer performs **risk checks**, **numeric validation**, and **timeline analysis**—delivering secure, explainable, and actionable assessments tailored for **reliable MDoNER decision-making**.

FOR TRAINING LLM THAT GIVES FINAL PREDICTIONS



FOR PROCESSING LARGE DPRS



FRAMEWORKS & TECHNOLOGIES

1. **Text Extractor (pdfplumber)**: Extract text and tables from DPR PDFs.
2. **GPT-4**: Analyze DPRs to generate structured JSON with costs, timelines, and risks for training.
3. **Local LLM (LLaMA 3 8B)**: Summarize large DPRs into concise 20–30 page summaries with data and tables.
4. **Hugging Face (DeBERTa-v3 / FLAN-T5)**: Fine-tune models to classify and score project risks.
5. **Transformers Library**: Optimize LLMs using **LoRA** and distributed training.
6. **JSON**: Store extracted text, summaries, and risk outputs.
7. **vLLM**: Run local LLMs efficiently with quantization for fast, memory-light inference.

FEASIBILITY

What make this idea realistic is that:

- Government DPRs follow **standardized structures**, enabling **consistent data extraction**.
- **Open-source LLMs, LoRA fine-tuning and OCR tools** can be customized for **domain-specific training and cost effectiveness**.
- Feasible integration with existing DPR management systems using **APIs and JSON formats** locally generated, thus also ensuring security.
- Our **modular pipeline (OCR → LLM → JSON → Dashboard)** can be built incrementally and validated at each stage

POTENTIAL CHALLENGES/RISKS

- Limited **availability of annotated DPR datasets** for supervised fine-tuning.
- Handling **document variations** (scanned images, formatting inconsistencies).
- High **computational requirements** for LLM training and inference.
- Model explainability and **bias detection** in AI-generated risk assessments.

PROPOSED SOLUTIONS

- Generate **synthetic DPRs**, use **GPT-assisted labeling**, and validate outputs through expert annotation cycles.
- Combine **OCR with layout-aware parsing tools** and implement adaptive preprocessing for diverse DPR formats.
- Utilize **parameter-efficient fine-tuning (LoRA/PEFT)** and **quantization**
- Two-tier LLM system will be trained in such a way that the **flags and risk factors are provided with justification**

POTENTIAL AND IMPACT OF THE PROJECT

- Enables faster and more reliable DPR assessments, **reducing manual workload for officials.**
- Provides **early detection of cost, schedule, and compliance risks** across development projects.
- Ensures **transparent, data-driven decision-making** for project approvals.
- Helps MDoNER **identify weak or inconsistent DPRs** before funding, improving efficiency of resource allocation.
- Creates a **scalable AI framework** adaptable for other ministries and government departments.

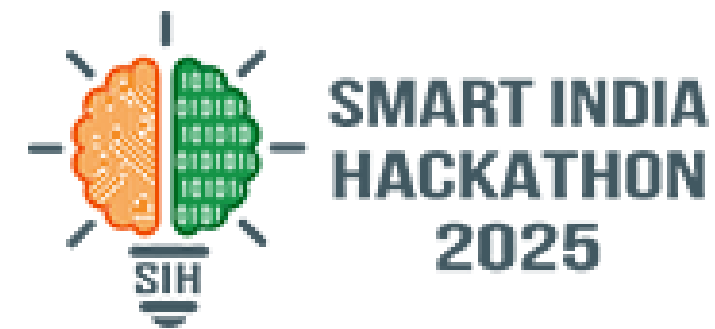


BENEFITS OF THE SOLUTION

- **Social benefit:** Accelerates project execution, fostering **regional development** and enhancing public welfare.
- **Economic benefit:** **Minimizes budget overruns** and resource inefficiencies via **early risk identification.**
- **Environmental benefit:** Enables prompt detection of projects lacking **necessary environmental clearances.**
- **Financial benefit:** Enhances **accountability, audit preparedness, and cost transparency** in DPR assessments.



RESEARCH AND REFERENCES



Documentation

[MDoNER – Guidelines for Preparation of DPRs](#)

[Hugging Face Transformers Documentation](#)

[Hugging Face PEFT](#)

[pdfplumber Library Docs](#)

[Llama documentation](#)

[vLLM: Easy, fast and cheap LLM Serving](#)

Research Papers and Literature

[Longformer: The Long-Document Transformer](#)
([Beltagy et al., 2020](#))

[Explainable AI for Public Sector Decision-Making](#)
([GovAI Report, 2023](#))

[Automated Risk Detection in Infrastructure](#)
[Project Proposals \(IEEE, 2022\)](#)

From Team NEXUS

THANK
YOU!